

ELYM 328 Laboratory

Experiment 1. Introduction to software development for microcontrollers

Aim: To give some tutorial on software development for microcontrollers using the PIC microcontroller family and PIC16F84 as a case study.

Equipment: Microchip MPLAB IDE software.

Reference materials: MPLAB IDE Quick Start Guide, PIC16F84A Data Sheet and PICmicro Mid-Range MCU Family Reference Manual.

INTRODUCTION

This laboratory session gives a brief tutorial on the use of MPLAB IDE for the PIC family of microcontrollers. The tutorial will cover the following topics:

- Running the MPLAB IDE
- Selecting the Device
- Creating the Project
- Setting up Language Tools
- Naming the Project
- Adding Files to the Project
- Building the Project
- Creating Code
- Testing Code with the Simulator

PROCEDURE

Refer to Chapter 2 of MPLAB IDE Quick Start Guide.

DISCUSSION

1. Describe the steps that you took to fix the error messages generated by the assembler.
2. Identify the function/purpose of each of the variables and keywords of the assembly programme considered during this lab session.
3. Give the reason(s) why error messages was initially generated for PORTC
4. By referring to PIC16F84A Data Sheet, find out and write on the purpose of PORTB and TRISB
5. In your own words, describe what you think the programme that was considered during this lab is doing.

NB:

1. The lab report should be written in a standard format consisting such things as: Title, Aims/objectives, Introduction, Equipment, Procedure, Results, Discussion and Conclusion.
2. You are required to attach the following to :
 - your raw programme (before debugging)
 - a file containing all the error messages that were generated the first time you attempted to build the programme
 - your debugged programme.